ECON 301.A TERM PROJECTS

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INTRODUCTION:

The primary goal of this term project is to understand the relationship between income per person and internet user rate. To this end, a research question is prepared. The reasons for selecting the data are explained by referencing relevant articles. The limitations of the analysis are explained. Descriptive statistics are prepared and analyzed for the data. A plot table is prepared in the context of income per person and internet user rate, and necessary explanations are provided. Regression analysis, like other data and tables, is prepared in R Studio, and the intercept, slope coefficient, R-squared, and residual standard error values are analyzed. In the second part of the project, to better understand the internet user rate, two new variables are added to the analysis: urban rate and education index. The reasons for selecting these two new data will be explained by referencing relevant articles. Finally, a multiple regression analysis is prepared, and the regression is analyzed in detail, similar to the simple regression analysis. Outliers are explained to better understand the analysis. The concepts of heteroscedasticity and multicollinearity are also analyzed. Finally, the term projects are completed with the conclusion section.

RESEARCH QUESTION:

> **"How much does a nation's per capita income account for cross-national differences in Internet usage rates?"**

SOURCES OF THE DATA:

Kaggle
https://www.kaggle.com/datasets/sansuthi/gapminder-internet
https://www.datapandas.org/ranking/education-rankings-by-country

SAMPLE SIZE: N= 183 (For simple regression)

UNITS OF OBSERVATIONS:
Country (each observation represents a single nation.)

VARIABLE Y: Internet user rate (This gives us the percentage of the population using the internet.)

the internet usage data are obtained from Gapminder and originally collected by the International Telecommunication Union (ITU), the United Nations' specialized agency for information and communication technologies. According to the ITU, an internet user is defined as a person who has been online at least once in the previous three months (Gapminder, 2024).

VARIABLE X: Income per person (Gross National Income per capita in dollar terms)

The income per person data (GDP per capita, PPP, inflation-adjusted) are obtained from Gapminder. According to Gapminder (n.d.), the indicator was updated to version 14 using interpolations for all missing years based on an assumption of constant growth rather than a constant absolute increase in income. This methodological adjustment provides a more realistic representation of income dynamics, particularly for fast-growing economies with limited historical data. Additionally, historical revisions were applied to selected countries, including China, the United Kingdom, Cambodia, Lebanon, Malta, and Greenland, to improve data consistency and accuracy (Gapminder, 2024).

WHY DID WE CHOOSE THIS DATA:

Internet use is increasing in the developing world. There are many reasons for this increase. As students and young people, we frequently use the internet both in our classes and in our personal lives. We were curious about the reasons for this increase in internet usage, which we use so frequently and has become central to our daily lives. As part of our Econometrics course, we suspected that this might be closely related to income per capita because providing and quickly accessing internet infrastructure requires a certain level of infrastructure and income. The Internet is still sold for a fee, and access to reliable internet is not readily available everywhere. Therefore, we used this data to investigate the relationship between income per capita and internet usage across countries because we believe there is a strong link between income and financial status. We also used articles to reinforce our thoughts and predictions. According to Rajagukguk et al. (2022), access to electricity, economic growth, and foreign direct investment had a positive association with internet use and inflation was negatively associated with internet use. In other words, there is a

relationship between internet user rate and the economy. According to Chinn & Fairlie (2007), unsurprisingly, we confirm the importance of *per capita* income in explaining the gap in computer and Internet use. This article explains the relationship between internet user rate and the economy, or in other words, income per person.

LIMITATION OF THE ANALYSIS:

1-) Omitted Variable Bias: While examining the impact of income per person on internet user rate and the relationship between the two, there are other factors affecting internet user rate, which constitute the limitation of our analysis. Other influencing factors could be education index or urban rate because these factors also affect the internet user rate.

2-) Measurement error: Measurement error can also be a limitation of the data because these calculations use data from a single year. This makes it difficult to understand the variables within the context of the process.

3-) Endogeneity problems: This problem could be one of the limitations of the analysis because income per person affects the internet user rate, but at the same time, the internet user rate can also affect income per person.

4-)Income inequality and distributional effects: Another factor is income inequality. Income inequality within a country isn't taken into account when calculating income per person. In other words, income in a country may be very high, but its distribution may be unequal, which could reduce internet access in a country with high incomes.

DESCRIPTIVE STATISTICS ANALYSIS:

```
> print(desc_stats)
           Variable       Mean          SD         Min         Max        Q25        Q50        Q75   N
1 Income per Person 8247.45756 12484.77078 103.7758572 81647.10003 752.250889 2549.55847 9334.45646 183
2 Internet Use Rate   35.23164    27.98099   0.2100663    95.63811   9.949296   31.00438   55.64642 183
>
```

In this part of the term project, the descriptive statistics prepared in R studio will be analyzed. Their sample size (n) is equal to 183.

According to the analysis, the average income per person is $8,247.45, and the average internet user rate is 35.23 percent. The standard deviation of the income per person is 12,484.77, but the standard deviation of the internet user rate is 27.98 The difference in standard deviations compared to mean indicates that the distribution of income per person and internet user rate is in a wide range.

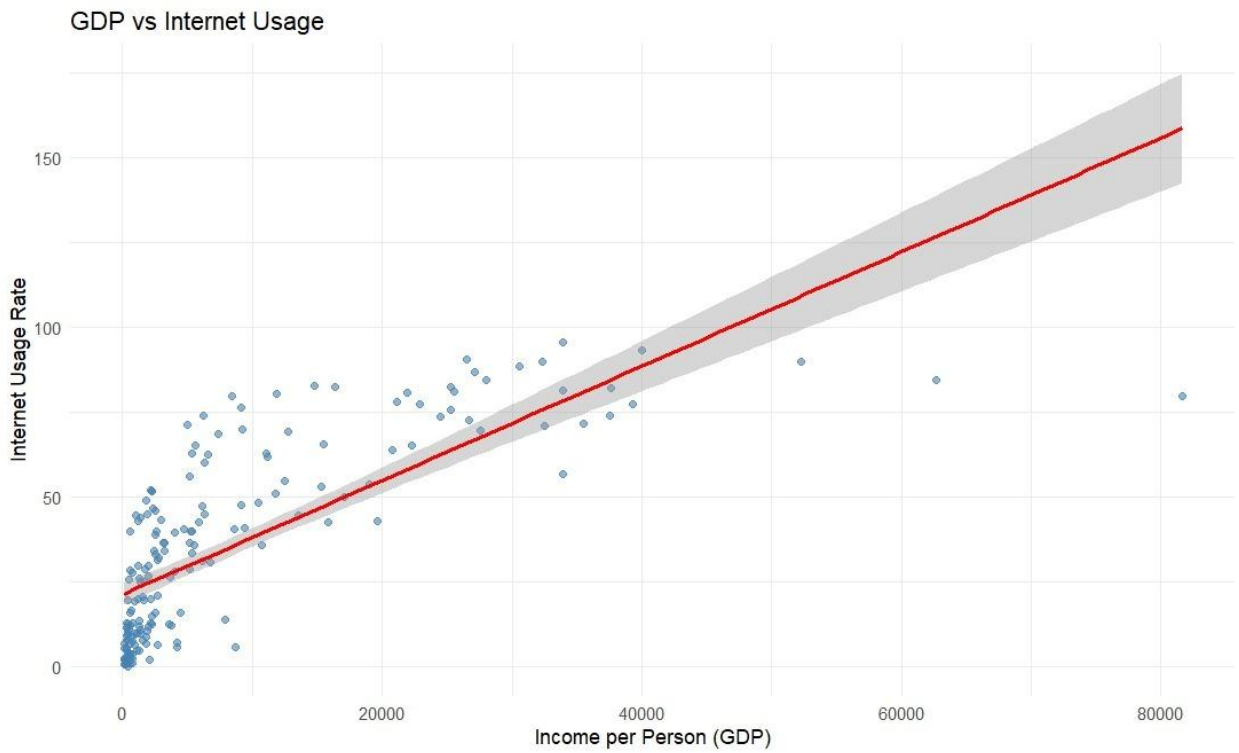
The minimum value for income per person is 103.78 and the maximum value is 81,647.100. This value shows that while income per person is even three digits in some countries, it reaches as high as 81,647.100 in others. In other words, these values indicate that income per person varies widely across countries, and this range is undeniably wide.

The minimum value for internet user rate is 0.21 and the max is 95.64. In other words, while the internet user rate is below 1 percent in some countries, it reaches a very high 95.64 percent in others. In fact, the difference between these two values reveals that internet usage rates are not distributed fairly across countries, just like income per person.

PLOT TABLE ANALYSIS

In the current part of the term project, will be analyzed the plot table which was obtained from R studio.

The plot table shows that there is a strikingly positive relationship between income per person and internet user rate. In other words, as income per person increases, the internet usage rate increases, or as it decreases, the income per person internet usage rate decreases. But of course, when looked at the chart in more detail, it can be seen that some countries with very low income per capita rates have higher internet usage rates than other countries with lower income per capita rates. But from a general perspective, we can say that as income increases, internet usage rates also increase.




```
Regression Equation: internetuserate = ????? + ?????*incomeperperson + u
> cat("Intercept (?????):", round(coef(model)[1], 4), "\n")
Intercept (?????): 21.3511
> cat("Slope (?????):", round(coef(model)[2], 6), "\n")
Slope (?????): 0.001683
> cat("R-squared:", round(model_summary$r.squared, 4), "\n")
R-squared: 0.5639
> cat("Adjusted R-squared:", round(model_summary$adj.r.squared, 4), "\n")
Adjusted R-squared: 0.5615
> cat("F-statistic:", round(model_summary$fstatistic[1], 2), "\n")
F-statistic: 234.05
> cat("P-value:", format.pval(pf(model_summary$fstatistic[1],
+                                model_summary$fstatistic[2],
+                                model_summary$fstatistic[3],
+                                lower.tail = FALSE)), "\n")
P-value: < 2.22e-16
```

REGRESSION ANALYSIS:

In this part of the term project, will be analyzed the regression analysis which was obtained from R studio.

x= income per person

y= internet user rate

y=21.3511 + 0.001683x

β_0=21.3511

β_0=intercept

This means that if the income per person is zero, the internet usage rate would be 21%. However, this is not economically feasible because the internet is a paid service, and accessing the internet without income is highly unlikely. Of course, since the internet is free in places like cafes and shopping malls, we still can't call it economically impossible.

β_1=0.001683

β_1= slope coefficient

This means that the internet usage rate rises by roughly 0.001683 percentage points for every dollar increase in per capita income. Thus, we can say that as income increases, internet usage rate also increases.

```
Residual standard error: 18.53 on 181 degrees of freedom
Multiple R-squared:  0.5639,    Adjusted R-squared:  0.5615
F-statistic: 234.1 on 1 and 181 DF,  p-value: < 2.2e-16
```

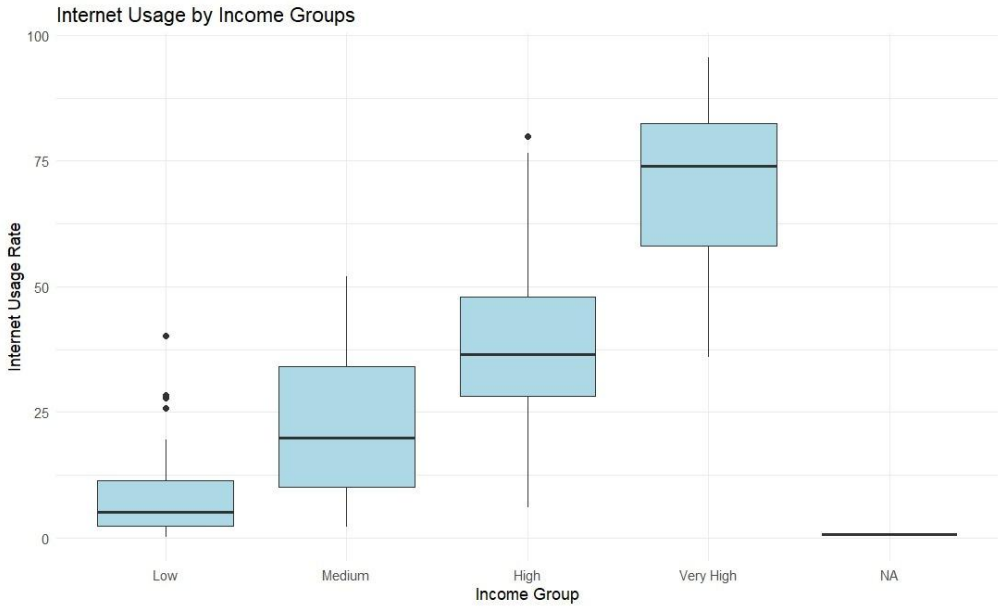
R-squared: 0.5639

The R-squared value of our model is nearly 0.5639, which means that about **56.39%** of the change in internet use rate can be explained by income per person.This is not a strong relationship with our variables. So we can say that other factors can affect the internet user rates such as education, government support, and technology infrastructure might also play an important role.

Residual Standard Errors (RSE): 18.53

Standard Error of the Regression: Based on the linear regression results that we can see above Residual Standard Error (RSE) is 18.53. We can interpret this value as the internet usage rate for a selected country deviates by approximately **18.53** percent. This value implies that there is a big unknown variability in internet usage

IN CONCLUSION OF THE FIRST PART OF THE TERM PROJECT:



Our primary aim with this initial econometric model was to explore the connection between a country's **economic capacity** and its level of **digital adoption**. Specifically, we tested whether higher **Income per Person** significantly affects the **Internet Use Rate**.

Key Takeaway from the Model

The results of our simple regression confirmed our initial hypothesis: there is a **positive and statistically significant relationship** between these two variables. The estimated equation, Y =21.3511 + 0.001683X , suggests that economic strength facilitates internet access and usage. To put the slope into perspective, a **1,000$ increase** in per capita income is

associated with an approximate **1.68 percentage point increase** in the internet usage rate. This proves that good internet access requires a strong national economy and stability.

However, the **R-squared value of 0.56** was a crucial point to consider. While we were able to explain almost a half of the variation in internet usage, the remaining variance clearly demonstrates that **income is only part of the story**. It's clear that other factors than economy must play a dominant role, which leads directly to our model's limitations.

Limitations and Critical Reflection

As a first-step simple linear model, our analysis has significant limitations that were crucial to identify:

- **We have a bias problem because we left out variables.** We know things like **literacy** and **government infrastructure** also affect internet use. By not including them, our model makes **Income per Person** look like it has a **bigger effect** on internet use than it really does.
- **Our income data is just an average, so it hides inequality.** A country can have a high average income, but if most of the money belongs to a few people, the average person can't afford internet access, making our prediction too high.
- **Correlation, Not Causation:** Because we used snapshot data, our model only shows that income and internet use are **related**, not that one definitely **causes** the other. Higher internet use might actually help increase income, and our simple model can't tell the difference.

SECOND PART OF THE TERM PROJECT:

Variable X2: Urban Rate: (Urban population (% of total) Urban population refers to people living in urban areas as defined by national statistical offices (calculated using World Bank population estimates and urban ratios from the United Nations World Urbanization Prospects))

It is considered that the urban rate is a factor affecting the internet user rate, just like the income per person, because the infrastructures in urban areas, such as fiber internet and cable connections, are a natural determinant of the internet user rate. Today, some countries have low internet user rates even with high incomes. This could be due to a variety of reasons. One of these reasons could be the low urban rate. If urban rate is included in the equation, structural differences between countries can be better understood. Another main reason for choosing the Urban rate is that the internet infrastructure in cities is more developed than in rural areas, because internet providers invest more in infrastructure in cities. Deep research was conducted to understand the impact of urban rates on internet user rates. According to Olson et al. (2025), because rural residents historically have lower rates of broadband internet access, they may have been less likely to conduct activities online than their urban counterparts, reflecting geographic digital inequalities. In other words, urban rate is a factor that affects internet access. According to Fang et al. (2024), indeed, digital inequality between urban and rural areas constitutes the most serious digital divide, stemming from longstanding conflicts and gaps between these areas. Rural areas, which are always at a disadvantage, lag behind their urban counterparts in digital race. So it can be said that living in a city or a rural area also affects internet usage rates.

Variable X3: Education index: (The Education Index, a composite indicator released by the United Nations Development Programme (UNDP) as part of its Human Development Reports, serves as the foundation for the education ranking data displayed on DataPandas. The Education Index combines expected years of schooling (EYS), which calculates the total number of years of schooling a child entering the educational system is anticipated to obtain, with mean years of schooling (MYS), which measures the average number of years of education completed by adults. Each nation receives a score between 0 and 1 that represents

its total level of educational attainment and potential once these factors are standardized and evenly weighted. As a result, the index offers a consistent, globally comparable gauge of educational advancement among nations.)

Another variable that affects the internet user rate is education, because as the level of education increases, skills such as adapting to technology and using technological devices increase. Another main reason for choosing the Education Index is that in countries where the level of education is high, individuals' desire to access information and conduct research increases, which is another factor affecting internet demand and internet user rate. Deep research was conducted to understand the impact of urban rates on internet user rates. According to Billon et al. (2021), distribution of formal education among citizens in a country is important in explaining Internet adoption. In other words, education is related to internet user rate.

Multiple Regression Analysis:

```
Call:
lm(formula = internetuserate ~ log(incomeperperson) + urbanrate +
    Education.Index.2022, data = final_data)

Residuals:
    Min      1Q  Median      3Q     Max
-36.686  -9.025   1.195  10.674  26.062

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)         -83.90521    5.70399 -14.710  < 2e-16 ***
log(incomeperperson) 10.34763    1.27645   8.107 1.31e-13 ***
urbanrate             0.01934    0.06236   0.310    0.757
Education.Index.2022 52.24016   10.55303   4.950 1.87e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 13.46 on 159 degrees of freedom
Multiple R-squared:  0.767,     Adjusted R-squared:  0.7626
F-statistic: 174.5 on 3 and 159 DF,  p-value: < 2.2e-16

>
```

A multiple regression model to see Income per person, Urban Rate, and Education collectively influence a country's internet usage rate.

Research Question:

> **"How much of the variation in internet user rates across countries can be explained by income per person, education index, and urban rate?"**

SAMPLE SIZE: N= 163

Y (Dependent Variable): Internet User Rate
Independent Variable X1: Log of Income per Person
Independent Variable X2: Urban Rate
Independent Variable X3: Education Index (2022)

Equation of the Multiple Regression Analysis:
$\hat{Y}=-83.91+10.35\ln(X1)+0.019(X2)+52.24(X3)$

Interpretation of the findings:

Intercept: β_0= -83.91

So this means that if all coefficients were zero the model would indicate that internet usage rate would be negative but this does not have any meaning in real life because in real life, the internet access rate cannot be negative. Additionally, in real life, it is unlikely that data such as income per person, urban rate, and education index will be zero.

Slope Coefficient: β_1= 10.35

So, this means that a 1 percent increase in income (log income) increases the internet user rate by about 0.1035 units. From this statement, it can be concluded that as income increases,

internet usage rate increases. Furthermore, P-value (1.31e-13) is statistically significant because P value is less than the alpha (0.05)

Slope Coefficient: β_2 = 0.019

So, this means that a 1 unit increase in urban rate increases the internet user rate by 0.019 units. However, P-value (0.757) is statistically insignificant because P-value is greater than the alpha (0.05). This shows that urban rate does not provide additional explanatory power to the model.

Slope Coefficient: β_3 = 52.24

So, this means that a 1 unit increase in the education index increases the internet use rate by approximately 52.24 units. From this statement it can be concluded that as education increases, internet user rate increases. Furthermore, P-value (1.087e-06) is statistically significant because P-value is less than the alpha (0.05).

R-Squared

Multiple R-squared: 0.767
Adjusted R-squared: 0.7626

When were compared multiple regression with simple regression we see that it improved hugely because in Simple Model R-Squared was **0.5639** so when we only look at income it only explains 56 percent of the internet usage variation but now we did the Multiple Regression Model we can explain almost 77 percent of the variation because R-Squared rose to 0.767. So just by adding the new coefficients Education and Urbanization we increased our explanatory power by 20 percentage points. This proves our conclusion in Simple Regression Model because as said Internet Usage can not be only affected by income alone.

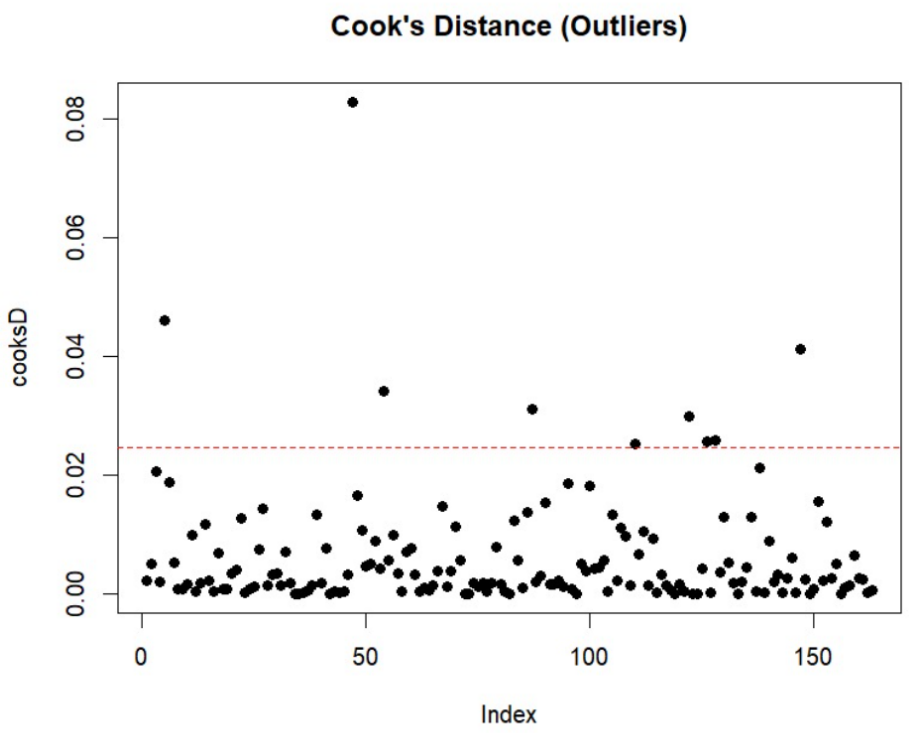
The adjusted R-square is 0.07626, which is slightly less than the R-square value of 0.767. This shows that the additional variables (urban rate and education index) meaningfully

contribute to explaining internet user rate. In other words, adding new values to the equation did not deteriorate the performance of the model, on the contrary, it increased its explanatory power.

Residual Standard Error: 13.46

The residual standard error (13.46) represents the typical size of the regression's prediction error. The typical size of the regression's prediction error is shown by the residual standard error (13.46). The residual standard error in the linear regression analysis is 18.53, but this number drops to 13.46 in the multiple regression. Based on this, it can be observed that the unexplained variation in the equation decreases with the addition of the urban rate and education index. However, after including the urban rate and education the model still leaves about 13.46 percentage points of unexplained variation in internet user rate.

Outliers:



This graph of Cook's Distances shows the Cook's Distance for each observation in the model of multiple linear regression. In order to find which of the influential outliers will cause significant change in the estimates of the regression coefficients upon deletion from the model, we need to use this type of graph.

Purpose: Cook's Distance shows how much the regression model is changed by deleting a single observation from the model. If an observation has a high Cook's Distance, then that observation is a high leverage and/or outlier, thus greatly influencing the results of our model.

Threshold: The dashed red line in this graph is an example of what many consider to be an upper limit (i.e., approximately .025) on where to look for influential points. Any observation that lies above this line may warrant closer inspection.

Identified Outliers & Points of Influence: It can be seen from the graph that several of the observations lie above the dashed red line, therefore these observations are considered influential outliers.

Most Influential Point: The single most influential point, located at Index 50, (50th country in our dataset), has the largest Cook's Distance, at approximately .08, and is the most influential observation in our model.

Detection (using residuals): The regression output provides the values for the minimum and maximum residuals; **Min=−36.686** and **Max=26.062**. The wide variability in the residuals suggest that there are possible outliers present.

Impact: Countries with a large negative residual (approximately −36.686) have an actual internet use rate that is far below what the model predicts. Conversely, countries with a large positive residual (about 26.062) have rates that exceed what was predicted by the model. If these points influence the slope estimates and the general fit of the model then the estimates will be biased.

Heteroscedasticity:

Definition: Heteroscedasticity happens when there are differences in the residual variances for each level of an independent variable. As long as the spread of residuals increases with higher values of income the model has heteroscedasticity.

Impact of Heteroscedasticity: Although Heteroscedasticity does not affect the estimated coefficients, it does invalidate the standard errors and p-values. Therefore, the t-statistics and F-statistics may also be incorrect and can lead to unreliable interpretations of statistical significance for your independent variables.

Multicollinearity:

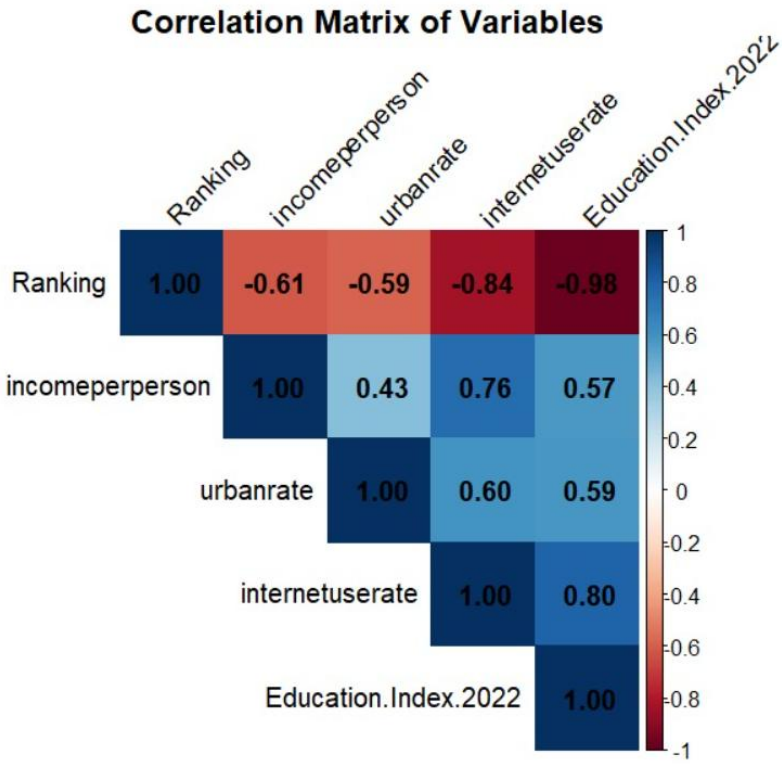
Definition: If there are high correlations between the independent variables (Urban Rate, Income per person, and Education Index), this will be referred to as multicollinearity. It would be expected to some extent because in many cases a country's income level correlates with how much of the population lives in urban areas and also correlates with the quality of its education system.

Possible Explanation: There could be a relationship between the non-statistical significance of Urban Rate, and it being very highly correlated with Income per Person and Education Index. When there is multicollinearity in a model, it makes it difficult to determine which of the variables that are strongly related to each other contribute uniquely to the dependent variable, and therefore the model can make it seem like a couple of variables (the ones with multicollinearity) are statistically insignificant when they really do contribute to the dependent variable.

Impact: The existence of multicollinearity causes both very large standard errors and small t-statistics for all of the variables involved in the multicollinearity, which makes it difficult to reject the null hypothesis for those variables. Therefore, even though a variable may have a true effect on the dependent variable, the inability to reject the null hypothesis indicates that we cannot conclude that the variable does have an effect. Since the model is estimating the overall fit of the data (R^2), the instability of the coefficients of the variables involved in multicollinearity does not affect the overall fit of the model.

Correlation Between Variables:

```
> print(round(cor_matrix, 2))
                    incomeperperson internetuserate urbanrate Ranking Education.Index.2022
incomeperperson                1.00            0.76      0.43   -0.61                0.57
internetuserate                0.76            1.00      0.60   -0.84                0.80
urbanrate                      0.43            0.60      1.00   -0.59                0.59
Ranking                       -0.61           -0.84     -0.59    1.00               -0.98
Education.Index.2022           0.57            0.80      0.59   -0.98                1.00
>
```



Multicollinearity occurs when the independent variables are highly correlated with each other (typically r > 0.70 or r > 0.80).

1- There are moderate positive inter-correlations among the independent variables of a multiple regression model (Income per Person, Urban Rate and Education Index) as indicated by the correlation matrix. The correlation coefficient between Income per Person and Urban Rate is 0.43, between Income per Person and the Education Index is 0.57 and between Urban Rate and the Education Index is 0.59.

2- The correlations among our three main predictors (Income, Urban Rate, and Education Index) are all below 0.60. This suggests that while they are related the correlation is not high enough to cause severe multicollinearity.

Standard vs Robust SE:

```
Regression Results: Standard vs Robust SE
=========================================================
                           Dependent variable:
                       ---------------------------------
                           internetuserate
                             Standard OLS       Robust SE
                                 (1)               (2)
---------------------------------------------------------
log(incomeperperson)        10.348***         10.348***
                             (1.276)           (1.271)

urbanrate                     0.019             0.019
                             (0.062)           (0.063)

Education.Index.2022        52.240***         52.240***
                             (10.553)          (10.846)

Constant                    -83.905***        -83.905***
                             (5.704)           (4.950)

---------------------------------------------------------
Observations                   163
R2                            0.767
Adjusted R2                   0.763
Residual Std. Error     13.458 (df = 159)
F Statistic          174.507*** (df = 3; 159)
=========================================================
Note:                         *p<0.1; **p<0.05; ***p<0.01
>
```

This table shows the data obtained from both standard OLS and robust (heteroskedasticity resistant) standard errors of the multiple regression analysis prepared to understand the variables affecting the internet user rate. In the table, there is very little difference between the OLS and Robust SE results. This data indicates that heteroskedasticity is not a major problem because Robust SE and OLS SE are very close to each other, in other words, there is no large fluctuation between errors. As a result, there is no significant problem in the form of the analysis.

Specific Analysis of Identified Outliers:

METHOD: Cook's Distance.

THRESHOLD: ~0.025 (Red dashed line).

TOP OUTLIER: Index 50 with a value of ~0.08.

RESIDUAL RANGE: -36.686 to 26.062.

Upon further investigation of the data points identified in our diagnostic plots, we determined the specific countries responsible for the most significant deviations. The observation with the highest Cook's Distance (approx. 0.08) and the largest negative residual (-36.69) corresponds to Equatorial Guinea. Despite having a relatively high income per capita, this nation's actual internet usage rate is far below what our model predicts based on its income, urbanization, and education levels. This suggests that in Equatorial Guinea, economic wealth has not translated into widespread digital adoption as expected. Conversely, the observation with the maximum positive residual (26.06) is Antigua and Barbuda. This country exhibits an internet usage rate that is approximately 26 percentage points higher than predicted, indicating that it has successfully fostered digital inclusion beyond what its economic and demographic variables would typically imply.

In conclusion:

In summary, this assignment provided us with valuable practical experience in identifying, estimating, and critically evaluating an econometric relationship. The results of the first analysis provide a solid starting point, but they clearly signal the necessity of moving toward a more sophisticated **multiple regression model** to capture the complexity of digital adoption globally. At the beginning of the term project, two new variables were added for the regression analysis: urban rate and education index. The reasons for choosing these variables were explained, and relevant articles were also used in this explanation, which can be found in the references section. Multiple regression analysis was then performed on these data, and the individual intercepts and slope coefficients were interpreted. The statistical significance of the data was analyzed using p-values. Then, the R square value and analysis were given and the residual error value and analysis were added to the assignment. Afterwards, Outliers,

Heteroscedasticity and Multicollinearity were also interpreted. Finally, Robust SE and OLS SE values were calculated and analyzed.

Resources:

Olson, K., Phillips, A., Smyth, J. D., & Stenger, R. (2025). The urban–rural digital divide in internet access and online activities during the COVID-19 pandemic. Rural Sociology, 90, e70012. https://doi.org/10.1111/ruso.70012

Billon, M., Crespo, J., & Lera-López, F. (2021). Do educational inequalities affect Internet use? An analysis for developed and developing countries. Telematics and Informatics, 58, 101521. https://doi.org/10.1016/j.tele.2020.101521

Fang, C., Chen, Z., Liao, X., Sun, B., & Meng, L. (2024). Urban–rural digitalization evolves from divide to inclusion: Empirical evidence from China. npj Urban Sustainability, 4, 51. https://doi.org/10.1038/s42949-024-00187-4

Rajagukguk, W., Kurniasih, N., & Siahaan, A. (2022). The demographic and economic features: The nexus with Internet use. Heliyon, 8(10), e11029.

https://www.sciencedirect.com/science/article/pii/S2405844022019740

Chinn, M. D., & Fairlie, R. W. (2007). The determinants of the global digital divide: A cross-country analysis of computer and Internet penetration. Oxford Economic Papers, 59(1), 16–44. https://doi.org/10.1093/oep/gpl024